

Nithin Jambula

Software Engineer | Robotics & AI Systems

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Summary

I'm a software engineer who enjoys writing the software that helps robots see, move and make decisions, and building AI systems that can search, reason and get real work done. I like understanding how the parts of a system fit together, from the first design sketch to something that runs reliably every day. Most of my work sits where robotics, AI and well-designed software meet, and I'm happiest turning a rough idea into something people can actually use.

Experience

Ananta Technologies

Dec 2025 – Present

Software Engineer – Robotics & Autonomy

India

- Own the camera-guided robot positioning module (AprilTag pose estimation and motion sequencing) as its main maintainer, keeping it reliable across multiple autonomous pilot programs.
- Rebuilt the communication layer for older CNC machines and H2D printers that ran closed software, and connected them to the in-house automation pipeline, removing manual operator steps on the print floor.
- Designed behavior trees and decision logic that turn mission plans into clear, step-by-step robot actions, with retries and fallbacks so robots recover from errors instead of stopping.
- Built and expanded a MuJoCo and Webots simulation test suite to 50+ scenarios covering motion, navigation and failure recovery, cutting validation time by 35%.
- Took 2 robots from simulation to the production floor, handling on-robot tuning, sim-to-real fixes and post-launch debugging.

AIR Centre, VIT-AP University

Jun 2024 – Nov 2025

Student Research Engineer – Applied AI & Security

Amaravati, India

- Built an AI framework that pairs LLM agents with rule-based reasoning to find security holes in web APIs automatically.
- Created an agent that reads live API details, writes attack tests (SQL injection, XSS, broken access control) and improves them after each failed attempt.
- Found 34% more vulnerabilities than standard scanners and cut manual testing effort by 70%; published at IEEE AIRC 2026.

Vulcan: Self-Driving Campus EV (VIT-AP Initiative)

Jan 2023 – Present

Software Lead – Perception & Navigation

Amaravati, India

- Lead the student team building perception, lane following and driving decisions for a self-driving electric campus vehicle.
- Trained a road-scene segmentation model (DeepLabV3+ on Mapillary) to 67% accuracy within the limited compute of an NVIDIA Jetson.
- Built lane-following and turn-handling control, and trained driving policies in the CARLA simulator with reinforcement learning (SAC) and imitation learning (DAgger).
- Showcased the working vehicle live at VLaunchpad 2025.

Sripto

Dec 2023 – May 2024

Machine Learning Engineer

Amaravati, India

- Built a document question-answering system (RAG) that answers questions from private company files accurately and in real time.
- Shipped AI tools including PDF chatbots, a product search matcher and a plain-English-to-SQL query engine.

Projects

PatchPilot: Multi-Agent Code Review System

Oct 2025

Python, LangGraph, LangChain, ChromaDB, Gemini, Groq, GitHub API | [GitHub](#)

- Built a code review system where separate AI agents check every GitHub pull request for security, code quality and logic issues, then a decision agent approves or blocks it based on risk.
- Designed the flow as a LangGraph state graph with conditional routing, so each pull request only goes through the checks it needs.
- Added long-term memory with a ChromaDB vector store so reviews learn from past issues, plus automatic patch suggestions and support for multiple LLM providers.

Context-Preserving RAG System for Enterprise Documents

Apr 2026

Python, FastAPI, Gemini, Embeddings, BM25, SQLite | [GitHub](#)

- Built a document question-answering system that gives source-backed answers from PDF, DOCX and text files.

- Combined meaning-based (vector) search with keyword (BM25) search, and added a query classifier that routes each question to the best search strategy.
- Used overlapping chunks to keep context across page breaks, and logged which passages were used for every answer so results can be traced and checked.

16fps: Multi-Agent Video Generation Platform

Oct 2025 – Present

Python, LangChain, Stable Video Diffusion, AnimateDiff, GCP

- Built a team of AI agents that turns a text prompt into a consistent video over a minute long.
- Agents plan scenes, remember characters, use tools and review their own output to fix mistakes, keeping the story and characters consistent at 16 FPS.
- Renders videos in parallel on zero-downtime cloud infrastructure; sold commercial rights and access to a media production company.

EchoSight: Navigation Glasses for the Visually Impaired (Patent Pending)

Dec 2024 – Present

YOLOv8, Raspberry Pi, OpenCV, Roboflow, GCP, Text-to-Speech | [Demo](#)

- Built smart glasses that detect obstacles and give spoken directions through Bluetooth earbuds.
- Fine-tuned YOLOv8 on 50+ custom object classes and optimised it to run at 45+ FPS on a Raspberry Pi.
- Awardee at the Vikas 2024 Innovation Challenge, showcased at inter-college innovation competitions, and filed for a patent.

AprilTag Docking Rover (ROS 2)

Dec 2025

Python, ROS 2, AprilTag, OpenCV, rosbag | [GitHub](#)

- Built a ROS 2 rover that finds and docks with its station on its own, using real-time AprilTag detection and pose estimation.
- Wrote a multi-stage mission controller (state machine) for approach, alignment and docking, with simulation worlds and rosbag-to-video tools for reviewing each run.

MatriAI: Semantic Profile Matching Service

Dec 2025

Python, FastAPI, LangChain, ChromaDB, MongoDB, Docker | [GitHub](#)

- Built a matching service that finds compatible profiles from a plain-English description, combining vector search with filters for age, location and preferences.
- Stored profiles in MongoDB and embeddings in ChromaDB, with batch indexing built to handle up to 300k profiles, and packaged everything with Docker Compose.

AI Voice Calling Agent

2024 – 2025

Python, Asterisk, ViciDial, Speech-to-Text, LLM, Text-to-Speech

- Extended an Asterisk and ViciDial calling setup into a voice agent that listens, understands and replies to callers in real time, with call routing and logging for review.
- Deployed it for live product outreach across different customer groups and demoed it end to end.

More Projects

github.com/nithin434

- **AI Sales Agent Platform:** FastAPI service that runs WhatsApp and LinkedIn reply bots and scores leads from 0 to 100 ([GitHub](#)).
- **WOAT:** WhatsApp assistant that learns your tone from past chats and replies in your style, built on Gemini ([GitHub](#)).
- **DeepShield:** deepfake video detector, +22% accuracy and 18% fewer false positives; published at IEEE iSAI-NLP 2025.

Publications, Patent & Awards

- **N. Jambula**, S. C. Sethuraman, et al., “A Neuro-Symbolic AI Framework for Adaptive and Explainable API Security Testing,” *7th International Conference on Artificial Intelligence, Robotics, and Control (AIRC)*, IEEE, 2026. DOI: [10.1109/AIRC69745.2026.11631384](https://doi.org/10.1109/AIRC69745.2026.11631384)
- **N. Jambula**, N. K. Reddy, K. Pandey, M. Mahesh, “Securing Reality: AI-Driven Detection of Deepfakes and Counterfeit Currency,” *20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, IEEE, 2025. DOI: [10.1109/ISAI-NLP66160.2025.11320726](https://doi.org/10.1109/ISAI-NLP66160.2025.11320726)
- **Patent Pending:** EchoSight, an AI-powered wearable navigation system for the visually impaired.
- **Awardee**, Vikas 2024 Innovation Challenge (EchoSight) | **Winner**, Engineering Clinics 2024 (real-time threat detection framework)

Skills

- **Languages:** Python, C++, Java, JavaScript, SQL, Bash
- **Robotics:** ROS 2, MuJoCo, Webots, CARLA, Behavior Trees, AprilTag, Visual Servoing, Sensor Fusion, Path Planning, Sim-to-Real, NVIDIA Jetson, Raspberry Pi

- **AI & LLM Systems:** LangGraph, LangChain, RAG, Hybrid Search (Vector + BM25), ChromaDB, Multi-Agent Systems, Tool Use, Agent Memory, Prompt Engineering, Hugging Face, Gemini, Voice AI
- **Services & Infrastructure:** FastAPI, Flask, Node.js, REST APIs, MongoDB, SQLite, Docker, Docker Compose, GCP, Linux, Git, CI/CD
- **Computer Vision & ML:** PyTorch, TensorFlow, OpenCV, YOLOv8, DeepLabV3+, ONNX, Reinforcement Learning (SAC, DAgger)

Certifications

- [Machine Learning Specialization](#) – Stanford & DeepLearning.AI | [Object Detection Bootcamp](#) – Google
- IBM: [Deep Learning Fundamentals](#), [Machine Learning with Python](#), [Machine Learning Specialist](#) – Advanced

Leadership

- **President**, [ACS International Student Chapter](#), VIT-AP Aug 2023 – Present
- **Marketing Member**, [SEDS VIT-AP](#) Jan 2023 – Present

Education

VIT-AP University , Amaravati	2023 – 2027
B.Tech in Computer Science and Engineering	CGPA: 8.0 / 10